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Muscle coactivation levels reflect a tradeoff between metabolic cost and performance when responding to physical disturbances during upper limb posture control.

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Abstract

Muscle coactivation refers to the simultaneous activity of muscles with opposing functions. Recent observations suggest that coactivation may improve performance by making the nervous system more responsive to sensory feedback. Understanding the relationship between the cost of coactivation and resulting performance can inform how the nervous system is able to use it to modify performance. This study will investigate the trade-off between metabolic cost and performance under different levels of muscle coactivation. Healthy participants will attempt to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances. The disturbances will rapidly extend or flex the elbow and displace the participant's arm from the target, requiring them to return to the target within 400 ms. Participants will first perform the task under self-selected levels of muscle coactivation, followed by biofeedback to adjust coactivation to [0.5, 2, 4, 8, 16]x their self-selected (1x) level. Metabolic cost will be measured with gas respirometry, muscle activity using surface EMG, and behavioral responses using success rates. Pilot data suggest that, at self-selected levels of coactivation, participants may achieve a success rate of 30-50%. It is anticipated that muscles will become more responsive to disturbances at higher coactivation levels, limiting displacement of the arm, improving success rates to nearly 100%, but increasing metabolic costs. These findings suggest that participants may prioritize metabolic cost over performance at self-selected levels of muscle coactivation. This research aims to offer insights into the role of muscle coactivation in responding to sensory feedback.

Literature Review

Introduction

Imagine a sprinter at the starting line of a competition; their muscles are tense and ready for action. This is an example of muscle coactivation, where the sprinter's hamstrings and quadriceps, two opposing muscles in their legs, cooperate to help control movement. It is widely believed that coactivation helps to stabilize our joints, control how stiff our limbs are, and fine-tune our movements, which are important aspects in sports but also in everyday life (Kesar et al., 2019; Vieira et al., 2019). Historically, coactivation's primary function was to make muscles stiff and resist movement (Vieira et al., 2019). However, recent studies have shown that there is more to coactivation especially in situations like sprinting where activating your opposing muscles seems to have a broader role in how we move (Kesar et al., 2019).

The connection between how well people perform and muscle coactivation is complex and not completely understood. More muscle coactivation generally means better joint stability, but it also uses more energy, which can impact performance (Molina-Rueda et al., 2023). In sports and rehabilitation, coactivation can help people make fewer mistakes, improve coordination, and respond more effectively to sensory feedback, such as proprioception and other sensory cues that guide movement and posture (Antohe et al., 2020). This literature review will look into what we know about coactivation, particularly how it works and why it matters in different situations such as sports or everyday activities. By looking at the balance between muscle coactivation, metabolic cost and performance we hope to shed some light on its complex role in human movement.

Physiological Mechanisms of Muscle Coactivation

Muscle coactivation is influenced by the control strategies of the nervous system and can vary in different levels of intensity or activation (Nagai et al., 2011). The control strategies include both the feedback and feedforward processes (Law & Avin, 2013). Traditionally, the feedforward mechanism prepares the body by activating the muscle for expected movements or perturbation, while the feedback mechanism involves fixing the muscle activity in response to sensory inputs such as an external force that causes a change in the position of the joint (Pienciak-Siewert et al., 2020). These mechanisms are very critical for maintaining human balance and doing precise movement, specifically in an unpredictable environment and have long been thought of as operating separately (Nagai et al., 2011). However, emerging research suggests that the relationship between feedforward and feedback mechanisms may not be as separate as once thought. While feedforward control usually prepares the body for expected movements or disruptions, it is becoming clearer that feedback mechanisms also play a role in adjusting muscle activity as changes happen in real time. In fact, instead of working independently, muscle coactivation and feedback seem to be connected, with the central nervous system (CNS) adjusting muscle activity based on both anticipated and actual disruptions. This idea challenges the belief that these mechanisms operate separately and shows that muscle control is more complex, especially in unpredictable situations.

Research has shown that the central nervous system (CNS) can control the level of coactivation based on our task demands, such as the need for stability or responding rapidly to mechanical perturbation (CF, 2017). For example, during a high-precision task our CNS increases our level of coactivation to minimize our movement to enhance our performance (Balshaw et al., 2018). To increase the level of muscle coactivation, the CNS utilizes a complex motor planning system that integrates sensory information and predicts the outcomes of the actions, which allows for an effective response to the environmental challenge (Law & Avin, 2013). Furthermore, research has shown that patterns of muscle coactivation can differ between populations. For example, older adults often show higher levels of muscle coactivation (Nagai et al., 2011). Rather than directly preventing the decline of muscle strength and coordination, this increased coactivation appears to be a compensatory strategy that helps offset the negative effects of age-related changes on performance (Nagai et al., 2011).

Muscle coactivation is also known to be organized into a group of muscles that can be activated together to produce a coordinated movement, which is known as muscle synergies (Kutch & Valero-Cuevas, 2012). Muscle synergy can help simplify the control of complex movements, and the nervous system will be able to coordinate actions more efficiently, by balancing the trade-off between performance and metabolic cost (Vieira et al., 2019). For example, studies related to electromyography (EMG) have shown different patterns of muscle activation which relate to various movements, such as reaching, grasping, and walking (Boudarham et al., 2016). These muscle synergies not only improve our motor performance but also improve the coordination of the muscle, which influences the overall energy expenditure (Le, 2023). Thus, understanding muscle synergies is very important to analyze how the CNS can control different levels of coactivation to achieve task-based goals, which can help in motor control and rehabilitation strategies (Le, 2023). For example, in a rehabilitation setting, understanding how muscle coactivation works can help create a medical intervention that can improve motor function in patients with an impairment related to the nervous system (Dyer et al., 2011).

Modulating Joint Stiffness Through Coactivation

A widely accepted idea is that the primary function of muscle coactivation is to control joint stiffness (Vieira et al., 2019). The body increases the stiffness of the joint through the coactivation of the agonist and antagonist muscles, which makes the joint more resistant to external perturbation (Gebel et al., 2019). Increasing the stiffness of the joint helps maintain joint stability, which is very helpful during tasks that apply a sudden force (Gebel et al., 2019). A person standing in a balanced position is an example of how coactivation of the ankle and knee muscles can help stabilize the body from swaying (Lévénez et al., 2005). A recent study has also shown that large levels of coactivation, which leads to a greater joint stiffness, reduce the risk of injuries during a dynamic activity, such as those involving the lower extremity, like walking (Molina-Rueda et al., 2023). This protective mechanism is primarily seen when the body has to respond to an unexpected lateral perturbation, increasing the overall stability of the joint (Vieira et al., 2019).

The role of muscle coactivation is seen across a wide range of motor tasks which includes maintaining the stability of the posture to executing rapid movements (Pienciak-Siewert et al., 2020). Muscle coactivation plays an important role in enhancing stability, optimizing responses, and improving movement precision (CF, 2017). The most popular role of muscle coactivation is associated with enhancing stability and balance (Deffeyes et al., 2012). For activities such as standing, walking, or carrying objects, maintaining balance requires a constant adjustment to the body's center of mass (COM). These adjustments to the body's COM are carried through muscle coactivation that provides a stable platform from which movements can be initiated (Deffeyes et al., 2012). For example, a person standing still is achieved by the coactivation of the calf and shin muscles to maintain ankle stability, reducing the risk of falling and swaying (Ting & Macpherson, 2005).

Exploring the Role of Muscle Coactivation in Sensory Feedback Processing

Many fine motor tasks, such as writing or working with small objects, require a high level of precision that can be achieved by higher levels of muscle coactivation (Law & Avin, 2013). A higher level of coactivation results in a reduction in the variability of movement and allows for more controlled movements because of increased joint stiffness (Balshaw et al., 2018). This reduction in variability is especially important in everyday situations where we need precise hand movements, like in surgery or when playing a musical instrument (Kutch & Valero-Cuevas, 2012).

While it is well established that muscle coactivation improves movement precision and stability, I am proposing something a bit different. Rather than focusing solely on its role in precision, I am interested in exploring how muscle coactivation could help the body process sensory feedback more efficiently. This idea, that muscle coactivation plays a role in sensory feedback processing, has not been widely discussed, which makes it a key part of my research. I believe that muscle coactivation not only helps with movement control but also prepares the body to respond more quickly to sensory signals, especially in situations that require rapid movements. For example, in sprinting, athletes coactivate their muscles before the starting signal, this prepares the muscles and nervous system for a quick and forceful start (Kesar et al., 2019). My project aims to see whether this muscle activation could also improve the body's ability to process sensory feedback in high-speed, dynamic movements.

Variability in Coactivation Across Individuals and Contexts

Muscle coactivation depends on multiple factors such as age, skill level, and the demands of the task (Nagai et al., 2011). Since coactivation differs across individuals and contexts, understanding this variation can provide knowledge about how controlling energy expenditure may impact the use of muscle coactivation as a strategy to process sensory feedback (Peterson & Martin, 2010). For example older adults show higher levels of muscle coactivation compared to younger adults for tasks that require balance and stability (Nagai et al., 2011). This increased coactivation serves as a mechanism to reduce the age-related decline in body coordination and muscle strength (Nagai et al., 2011). Although this increased muscle coactivation also results in increased metabolic cost, which can cause fatigue and reduced endurance in older adults (Peterson & Martin, 2010). Age can therefore significantly affect the extent and pattern of muscle coactivation.

A person skilled in a particular activity can influence muscle coactivation patterns. Professional athletes display coactivation strategies that reduce unnecessary muscle activity while maximizing performance (Balshaw et al., 2018). An experienced pianist for example uses minimal coactivation during performance to allow for greater fluidity in their movement (Pereira & Gonçalves, 2011). On the other hand, a beginner playing the piano shows a higher level of coactivation as they try to stabilize their movements and stay responsive to sensory feedback, correcting deviations before they become errors (Pääsuke et al., 2000). The task demands also play an important role in controlling the level of muscle coactivation. For tasks that need a lot of stability, like lifting a heavy object or keeping a steady posture, higher levels of muscle coactivation are used (Peterson & Martin, 2010). In contrast, tasks that are dependent on speed and fluidity may need lower coactivation levels to allow for more range of motion (ROM) and flexibility (Peterson & Martin, 2010). This variation in coactivation based on the task shows how the nervous system adapts to maximize performance.

Balancing Performance and Metabolic Cost

Muscle coactivation plays a pivotal role in affecting metabolic cost. While it can boost our performance by improving stability and responsiveness, it is generally understood to raise the body's energy demand (Peterson & Martin, 2010). Coactivation leads to more muscle activity, which means the body uses more energy (Peterson & Martin, 2010). This increase in energy demand can be measured using methods like gas respirometry, which looks at how much oxygen we consume and how much carbon dioxide we produce (Peterson & Martin, 2010). While existing studies suggest that tasks requiring more muscle coactivation, such as holding a steady posture against a force, may consume more energy, the exact relationship between coactivation levels, metabolic energy expenditure, and behavioral performance is still unclear (Peterson & Martin, 2010).

It is generally believed that people try to balance the energy cost of muscle coactivation with the benefits of improved performance and stability. For example, pilot data suggests that people tend to increase their coactivation to respond faster to visual stimuli, even if it means using more energy. On the other hand, for tasks that do not require that much performance, they might lower their coactivation to conserve energy. But right now, there’s no concrete evidence that directly explores this balance between coactivation, energy use, and performance. This is the objective of my project: to investigate the impact of coactivation on performance (e.g., success rates, speed) in dynamic tasks and to assess the increase in metabolic cost with varying levels of muscle coactivation. Ultimately, my goal is to better understand how people decide how much coactivation to use based on the task demands and the energy involved.

Interplay Between Muscle Coactivation and Sensory Feedback

Muscle coactivation might play a significant role in controlling neural sensory feedback, which is something I am interested in investigating further and possibly answering. Recent studies suggest that increasing muscle coactivation enhances how the CNS responds to external disturbances. For instance, Kesar et al. (2019) discovered that the excitability of corticomotor pathways for ankle muscles increases when there is coactivation between agonist and antagonist muscles. Increased excitability or responsiveness helps maintain balance and stability during dynamic activities (Boudarham et al., 2016). However, the specific role of coactivation in preparing the CNS for quick reactions during tasks requiring rapid responses to physical disturbances is not fully understood and has not been directly examined in empirical studies. The potential for the CNS to regulate coactivation levels based on task demands suggests that this relationship is crucial for motor control, yet it remains largely unexplored, which is what makes this research exciting.

Moreover, the concept of muscle coactivation as a compensatory mechanism is particularly important for people with motor control problems. Looking at people with multiple sclerosis, which is a progressive neurodegenerative disease of the CNS that is caused by axonal degeneration and demyelination, Molina-Rueda et al (2023) found that they show high levels of coactivation to maintain the stability during gait. This suggests that coactivation of the lower limbs during gait for patients with multiple sclerosis can result in increased joint stiffness and help to counteract the effect of external perturbations (Boudarham et al., 2016). As increased coactivation improves joint stability, it also increases metabolic costs, which emphasizes a trade-off between performance and energy expenditure (Peterson & Martin, 2010). Understanding coactivation patterns in various populations can help us learn how the brain processes sensory information and controls movement.

While different research studies on the primary motor cortex explains the neural mechanisms behind muscle coactivation, the evidence supporting how coactivation influences sensory feedback and neural processing is still limited. For instance, Balshaw et al. (2018) describe that different patterns of muscle coactivation correspond to specific motor tasks, suggesting that the CNS organizes muscle use and coordination based on the task at hand. However, the specific neural structures supporting coactivation and how they influence sensory feedback processing remain underexplored. It is also unclear how coactivation might change the processing of even basic neural circuits. Current research points to the potential for the CNS to adjust coactivation levels to enhance performance while minimizing energy use (Gebel et al., 2019), but this understanding is still in its early stages. Understanding how muscle coactivation interacts with sensory feedback could offer new insights, particularly in rehabilitation settings where the goal is to restore motor skills (Kutch & Valero-Cuevas, 2012).

Implications for Rehabilitation and Clinical Applications

Muscle coactivation has important uses in rehabilitation and medical settings. Some conditions, like muscle spasticity, hyperreflexia, and motor problems, show unusual muscle coactivation patterns (Dyer et al., 2011). Understanding these patterns helps doctors create treatments to improve motor skills and reduce abnormal muscle activity (Dyer et al., 2011). For example, muscle spasticity means increased muscle tone and reflexes (Dyer et al., 2011). This happens because of too much muscle coactivation, which makes movement hard and causes joint stiffness (Dyer et al., 2011). Treatments like botulinum toxin injections, physical therapy, and neuromodulation are used to lower the excessive coactivation and help improve motor function (Dyer et al., 2011).

Patients with motor impairments which includes stroke survivors show patterns of coactivation that may be disrupted (Dyer et al., 2011). This can lead to difficulties in movement coordination (Dyer et al., 2011). So rehabilitation programs that specifically focus on fixing the coactivation patterns can help fix movement patterns in these patients (Dyer et al., 2011). Some techniques such as biofeedback control, strength training and task-specific exercise can be used to change the coactivation level and improve functional outcomes (Dyer et al., 2011). Finally, understanding the neural mechanism related to coactivation can improve the development of more effective rehabilitation strategies.

Future Research Directions

Studying muscle coactivation is an area that's always changing, with lots of chances for future research. There are new ways to look at how the brain and body work with coactivation, like using advanced imaging and neuromodulation techniques (Pereira & Gonçalves, 2011). In order to understand the range of scenarios in which coactivation can be useful, research should focus on a larger of populations, which includes older adults, athletes, and persons with impairments. Also, looking into how coactivation relates to things like muscle fatigue can help us understand motor control better.

Conclusion

Muscle coactivation is important for controlling our movement which helps with stability, accuracy, and responsiveness across many tasks. The brain controls this by complex interactions between feedforward and feedback processes, adjusting coactivation levels based on what the task needs and the person’s abilities. In conclusion, understanding muscle coactivation is important for maximizing our performance, managing energy expenditure, and creating better rehab strategies for people with movement issues.

Question

How does muscle coactivation affect the trade-off between performance and metabolic cost during upper limb posture control with mechanical disturbances?

Hypothesis

There is a trade-off between muscle coactivation, metabolic cost, and task performance.

Rationale

This project hopes to find some connection between muscle coactivation, performance, and metabolic cost, addressing an important gap in current research. While we know that muscle coactivation helps stabilize joints and enhance movement, we still do not have a complete understanding of how it affects metabolic cost, especially in tasks involving upper limb posture control. Additionally, there is limited research on how coactivation impacts fast, precise tasks that demand quick adjustments. The aim of this study is to gain a deeper understanding of how coactivation influences performance, including success rates and speed, in dynamic tasks, and to examine how it affects metabolic cost at varying levels of coactivation. The prediction is that higher levels of coactivation will improve performance, such as reducing error rates, but will also result in increased metabolic cost. This gap is particularly significant because the central nervous system adjusts coactivation levels based on task demands, especially in dynamic environments. By tackling this issue, the project will let us know how different levels of coactivation influences performance and energy use (metabolic cost). These findings could have real-world applications in rehabilitation, especially for individuals with motor impairments, where it is important to improve motor function while also managing energy use. Finally, this research has the potential to influence clinical interventions and enhance our understanding of motor control, making the project timely and necessary for both performance optimization and rehabilitation.

Methodology

Kinarm Exoskeleton Adjustment:

Healthy participants were asked to place their arms in the troughs of the Kinarm exoskeleton robot which limited movement in a single plane, allowing only horizontal motions. We adjusted the device to ensure that participants could move smoothly without restrictions. We also adjusted the chair height so their arms were level with the exoskeleton arms, with feet flat on the footrest and back supported. The shoulder indicator light was turned on to ensure their shoulders were level, and aligned with the acromion process. The armrests were fine-tuned to support their arms comfortably, with elbows slightly bent, and aligned the exoskeleton arms with the participant's arm length to ensure the joints matched the shoulder, elbow, and wrist. In addition, the Kinarm device was connected to a TV screen that displayed the targets. We closed a shield so participants could not see their hands, only the visual representation of their hands as a white cursor on the screen.

Participant Preparation

Muscle activity was measured using EMG electrodes placed on specific muscles. If necessary, we shaved small areas using disposable razors to improve the signal quality. We cleaned the muscles with an alcohol swab and a reference electrode was placed on the lateral epicondyle while the EMG electrodes were placed on the brachioradialis, triceps lateralis, biceps (short) and triceps (longus) muscles. If it was their first time, we created a new subject profile in the EMG computer and recorded their age, height, and weight before fitting the VO2 mask. Participants also wore a VO2 mask which covered their nose and mouth and this measured their metabolic activity while performing the task. They were asked to minimize speaking during the task to avoid affecting their breathing rate. The machine was calibrated, and participants were informed that they would only see the white cursor aligned with their index finger after calibration.

First Task Protocol:

Participants were told that a small target would appear at the center of the screen, and they needed to move the white cursor representing their hand to the center of the target. They were asked to maintain a fixed arm posture while the robot introduced a mechanical disturbance that rapidly extended or flexed the participants' elbows. This displaced their arm from the center of the target position and participants were required to return their arm to the target within 400 milliseconds. Feedback was provided based on their performance: a green target indicated success, while a blue target indicated they were too slow or inaccurate. After each task, participants were given a brief break. During the experiment, we recorded the time and reference EMG values, using the first condition to determine the reference EMG values for the biceps with MATLAB.

Second Task Protocol:

For the second task, participants were instructed to bring the white cursor into the center of a circular target. A slider (magenta in color) and biofeedback target (green rectangular box beside the circular target) were also displayed. They were told to match the slider, representing muscle activation, to the biofeedback target, which showed the required muscle activation for that trial. The machine would then apply a force, and participants had to return the cursor to the center of the circular target. If the hand get back into the centre of the circular target within the time limit, then the target would appear green and if it was done within time, then the target would appear blue. This task was performed until failure.

Final Task:

In the final task, lasting about 2 minutes, participants were instructed to bring the white cursor into the center of the circular target and hold that position while relaxing as much as possible, even as a small load was applied. After the task, we saved the data, and participants were compensated with $20.

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